

大学物理课后练习十四答案

一、选择题

1. (A)

解：将介质板插入 C_1 后， C_1 值增大，电压 U_1 减小，电压 U_2 增大，故 C_1 上电量增大。

2. (A)

解：在球心 O 为圆心，以半径 r ($r > R$) 作高斯面 S，由电介质中的高斯定理 $\int_q \vec{D} \cdot d\vec{S} = q$,

$$D \cdot 4\pi r^2 = q, \quad E = \frac{q}{4\pi\epsilon r^2} \quad V_A = \int \vec{E} \cdot d\vec{l} = \int_R^\infty dr = \frac{q}{4\pi\epsilon R}$$

3. (B)

解：均匀带电球体和球面所带电量相同，故 $r > R$ 时，电场强度相同，电场能相等，但均匀带电球体内部场强不为零，故球体电场能大于球面电场能。

4. (C)

解：两者用导线连接前，系统电场能为空腔内外电场能之和，当将两导体用导线连接后，电荷集中于外表面，空腔导体内场强为零，空腔外场强不变电场能不变，故系统电场能将减少。

二、填空题

1. 增加；减少

解：将电容器充电后， $U_1 = U_2 = U, Q = Q_1 + Q_2, \frac{C_1}{C_2} = \frac{Q_1}{Q_2}$ ，将介质板插入 C_1 后， C_1 增大，

则 Q_1 增加， Q_2 减少。

$$2. \quad D = \frac{\lambda}{2\pi r}; \quad E = \frac{\lambda}{2\pi\epsilon_0\epsilon_r r}; \quad C = \frac{2\pi\epsilon_0\epsilon_r}{\ln \frac{R_2}{R_1}}; \quad W_e = \frac{\lambda^2}{4\pi\epsilon_0\epsilon_r} \ln \frac{R_2}{R_1}$$

3. 减少；增加

解：图 a 中，介质板拉出前后，U 不变，则 E 不变，由 $W_e = \frac{1}{2} \epsilon_0 \epsilon_r E^2$ ，故静电能减少。

图 b 中，介质板拉出前后，Q 不变，而 C 减小，由 $W_e = \frac{1}{2} \frac{Q^2}{C}$ ，故静电能增加。

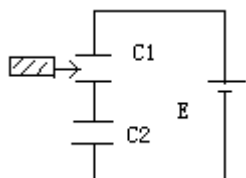


图 (1)

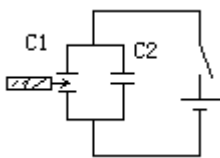


图 (2)

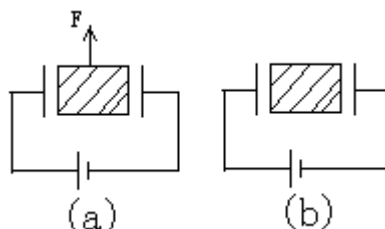


图 (3)

$$4. \quad \Delta W_e = -\frac{Q^2}{4C}$$

解：并联前 $W_{e1} = \frac{1}{2} \frac{Q^2}{C}$, $W_{e2} = \frac{1}{2} \frac{4Q^2}{C}$, 总电场能为 $W_{e1} + W_{e2}$, 并联后电场能 $W'_e = \frac{9Q^2}{2C}$,

$$\text{故 } \Delta W_e = \frac{9Q^2}{2C} - \left(\frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} \frac{4Q^2}{C} \right) = -\frac{Q^2}{4C}$$

$$5. \Delta W_e = -\frac{\varepsilon_r - 1}{\varepsilon_r} W_e$$

解：将带电金属球浸没在相对介电常数为 ε_r 的电介质中，其球外场强

$$E = E_0 - E' = E_0 - \frac{\varepsilon_r - 1}{\varepsilon_r} E_0 = E_0 \left(1 - \frac{\varepsilon_r - 1}{\varepsilon_r} \right), \text{ 故其静电场能的增量为 } \Delta W_e = -\frac{\varepsilon_r - 1}{\varepsilon_r} W_e$$

$$6. W = \frac{1}{2} \varepsilon_0 \varepsilon_r \left(\frac{U_{12}}{d} \right)^2$$

三、计算题

1. 解：(1) $C_{AC} = 4 + 8 = 12$, $C_{CD} = 6 + 2 = 8$

$$\frac{1}{C_{AB}} = \frac{1}{C_{AC}} + \frac{1}{C_{CD}} + \frac{1}{C_{DB}} = \frac{1}{4}, \quad C_{AB} = 4$$

$$(2) Q = C_{AB} U = 48$$

$$U_{AC} = \frac{Q}{C_{AC}} = 4, \quad U_{CD} = \frac{Q}{C_{CD}} = 6, \quad U_{DB} = \frac{Q}{C_{DB}} = 2$$

2. 解：设电容器极板上电荷面密度为 $\pm\sigma$

$$(1) \text{ 由电介质中的高斯定理, 知 } D_1 = D_2 = \sigma; \quad E_1 = \frac{\sigma}{\varepsilon_0 \varepsilon_{r1}}; \quad E_2 = \frac{\sigma}{\varepsilon_0 \varepsilon_{r2}}$$

$$\Delta V = E_1 d_1 + E_2 d_2 = \frac{\sigma}{\varepsilon_0} \left(\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} \right)$$

$$C = \frac{Q}{\Delta V} = \frac{\sigma S}{\frac{\sigma}{\varepsilon_0} \left(\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} \right)} = \frac{\varepsilon_0 S}{\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}}} = 9.32 \times 10^{-11} F$$

$$(2) \sigma = \frac{Q}{S} = \frac{C \Delta V}{S} = 1.77 \times 10^{-5} C/m^2$$

$$(3) D_1 = D_2 = \sigma = 1.77 \times 10^{-5} C/m^2; \quad E_1 = \frac{\sigma}{\varepsilon_0 \varepsilon_{r1}} = 4.0 \times 10^5 V/m; \quad E_2 = \frac{\sigma}{\varepsilon_0 \varepsilon_{r2}} = 1.0 \times 10^6 V/m$$

3. 解: (1) 电荷分布如图所示, 电场存在于 I 和 II 两个空间内, 场强为 $E = \frac{Q}{4\pi\epsilon_0 r^2}$, 则

$$W = \int_{V_1} \frac{1}{2} \epsilon_0 E^2 dV + \int_{V_2} \frac{1}{2} \epsilon_0 E^2 dV = \frac{Q^2}{32\pi^2 \epsilon_0} \left(\int_{R_1}^{R_2} \frac{4\pi r^2 dr}{r^4} + \int_{R_3}^{\infty} \frac{4\pi r^2 dr}{r^4} \right) = \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3} \right) = 1.82 \times 10^{-4} J$$

把系统看成两个电容器: 同心球电容器和孤立电容器, 则 $W = W_1 + W_2 = \frac{Q^2}{2C_1} + \frac{Q^2}{2C_2}$

$$C_1 = 4\pi\epsilon_0 \frac{R_2 R_1}{R_2 - R_1}, \quad C_2 = 4\pi\epsilon_0 R_3, \quad \text{则 } W = 1.01 \times 10^{-4} + 0.81 \times 10^{-4} = 1.82 \times 10^{-4} (J)$$

(2) 此时电荷只分布在外壳, I 腔内 $E=0$, 电场仅存于 II 空间内, 场强仍为 $E = \frac{Q}{4\pi\epsilon_0 r^2}$,

$$\text{所以 } W' = \frac{1}{2} \int_{II} \frac{1}{2} \epsilon_0 E^2 dV = \frac{Q^2}{32\pi^2 \epsilon_0} \int_{R_3}^{\infty} \frac{4\pi r^2 dr}{r^4} = \frac{Q^2}{8\pi\epsilon_0} \cdot \frac{1}{R_3} = 0.81 \times 10^{-4} J$$

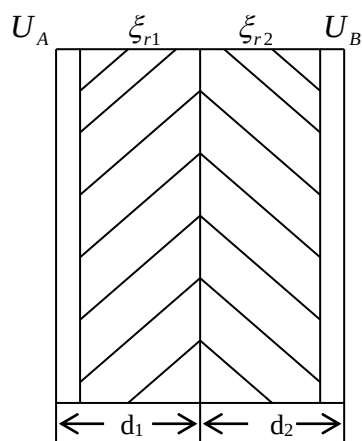


图 (4)

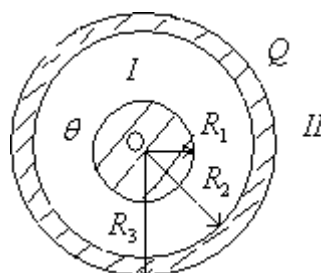


图 (5)

4. 解: $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 S}$

$$\Delta V = Sd$$

$$w_e = \frac{1}{2} \epsilon_0 E^2$$

$$\Delta W_e = w_e \Delta V = \frac{Q^2 d}{2\epsilon_0 S}$$